UNIT-4

Computer Graphics

✓ B.Tech
✓ Aktu
→ numericals will be provided on the numerical video.

Syllabus :-

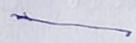
Curves and surfaces :- Quadratic surfaces, sphere, Ellipsoid, Blobby objects, Introductory concepts of spline, Bspline and Bezeir curves and surfaces.

Curves :-

A curve is a smooth, continuous line formed by connecting many points.



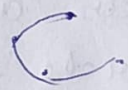
Curves are used to represent round, smooth, and shapes that cannot be drawn properly using straight lines.



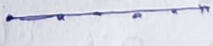
Ex from real life :-

- Road turns
- Rainbow
- car body design





- curves are used in :-
  1.) Computer graphics
  2.) Animation.
  3.) Game design



curvature = 0

Note :-
A line is considered a type of curve in mathematics and computer graphics.
It has zero curvature.

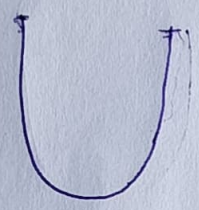
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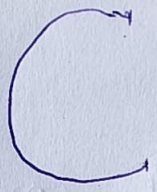
Types of curve :-

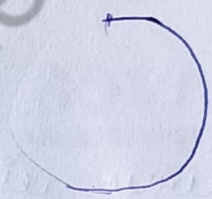
Curves can be classified based on their shape and how their ends meet.

1.) Open curve :-

- An open curve is a curve whose starting point and ending points are different.
- Ends do not meet.
- Does not enclose an area.
- Ex :- A road from one city to another

1.) 

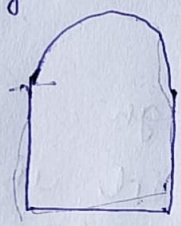
2.) 

3.) 

2.) Closed curve :-

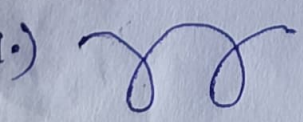
- A closed curve is a curve whose starting point and ending point are the same.
- Ends meet to form a loop.
- Encloses an area.
- Ex :- Circle, Rectangle.

1.) 

2.) 

3.) Crossing curve :-

- A crossing curve is a curve that intersects or crosses itself at one or more points.
- May be open or closed.
- The curved overlaps or crosses itself
- Crossed wires or tangled string.

1.) 

2.) 

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Representation of a curve :-

To draw, store and manipulates curves on a computer, they must be represented in a mathematical order form.
This is called curve representation.

* Types of curve representation :-
curves are mainly represented in three ways :
1.) Implicit representation.
2.) Explicit representation
3.) Parametric representation

1.) Implicit Representation :-

In implicit representation, a curve is defined by an equation of the form :

$$f(x,y) = 0$$

The curve consists of all points (x,y) that satisfy this eqn.



Ex :- Circle :- $x^2 + y^2 - r^2 = 0$.

- It is easy to test whether a point lies on the curve.
- It is not suitable for animation.

2.) Explicit Representation :-

In explicit representation, one variable is expressed directly in terms of another.

$$y = f(x)$$

Here, y depends on x.

Ex :- Parabola : $y = x^2$.

- Easy to compute y for a given x.

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3.) Parametric representation :-

In parametric representation, both x and y are expressed in terms of a third variable called the parameter (t)

$$x = f(t)$$
$$y = g(t)$$

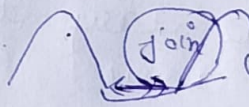
- It is most flexible and powerful method.
- Can represent all types of curves.
- It is most commonly used in computer graphics.

Parametric & Geometric continuity :-

* Continuity of curves :-

When two curve segments are joined together, continuity describes how smoothly they are connected.

In computer graphics, continuity is mainly of two types :-



1.) Parametric continuity
2.) Geometric continuity.

1.) Parametric continuity :-

Parametric continuity depends on the parameter values of the curves.

It ensures that the curve and its derivatives are continuous at the joining point.

Types of Parametric continuity :-

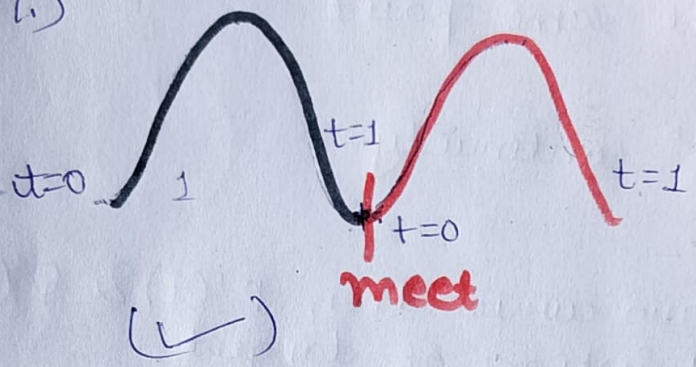
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a) C^0 continuity :-

- The end point of the first curve meets the start point of the second curve.
- curves are connected but may have a sharp corner.

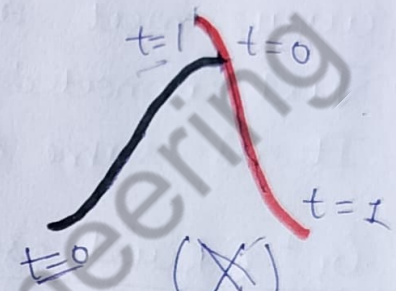
→ Just connected, not smooth.

1.)



(✓)

2.)



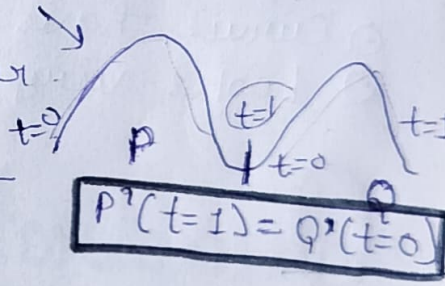
(X)

not C^0 continuity

b) C^1 continuity (first order) :-

- End points are same. ✓
- first derivatives (direction and speed) are equal.

→ smooth turn, no sharp corner.



c) C^2 continuity :- (second order) :-

- The curves meet, direction is same and curvature is also smooth.
- Ex :- car body design.

if $P'(t=1) = Q'(t=0)$ & $P''(t=1) = Q''(t=0)$

2.) Geometric continuity :-

Geometric continuity checks how the curve looks, not how fast it is drawn.

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In simple :-

Parametric continuity → maths + speed matters (→ mathematical)
Geometric continuity → shape + look matters.
↓
visual.

a) G^0 (zero order) :-

- curves meet at the same point.

→ Just connected

- It is same as C^0 continuity.

b) G^1 (first order) :-

- if P and Q are two curves then P'(t) & Q'(t) must have same direction of tangent.
- curves have the same direction, but speed can be different.

c) G^2 continuity :-

- Curves have the same curvature.
- Looks very smooth.

Quadratic surfaces :-

Quadratic surfaces are 3-D shapes that are formed using second-degree equations (power 2) of x, y and z.

→ If the highest power of x, y or z is 2, the surface is called a quadratic surface.

The general eqn is :-

$Ax^2 + Bx + C$ — quad. eqn — 3-D shape

$9x^2 + 5x + C$

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* General eqn of a quadratic surface :-

$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$

This eqn represents diff types of curved surfaces depending on the coefficients.

→ Common types of quadratic eqn surfaces :-

1.) Sphere :-

- All points are at a constant distance from the center.

$$x^2 + y^2 + z^2 + r^2 = 0$$

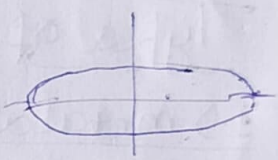


2.) Ellipsoid :-

An ellipsoid is a 3D quadratic surface that looks like a stretched or compressed sphere.

Eqn :-

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$



If $a = b = c$, the ellipsoid becomes a sphere.

Blobby objects :-

- Blobby objects are soft and smooth 3D shapes used in computer graphics.
- They look like drops of liquid or clay that join together when they come close.

Ex :-

Imagine water drops :-

- When drops are far apart → they look separ
- When drops come close → they merge smoothl into one shape.

Blobby objects work in the same way

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They are mainly used to model soft, natural shapes like liquids, clouds.

Mathematically concept :-

The object surface is defined by :-

$$\sum_{i=1}^{n} f_i(x, y, z) = T$$

where :

f_i = field function of each metaball.
T = Threshold function.

* Why they are called "Blobby" :-

Because they look like blobs of liquid that flow and stick together naturally. ⟶

Types of Blobby objects :-

1.) Metaballs :-

- made of spherical fields.
- when spheres come close, they blend smoothly.

2.) Skeleton :-

- Instead of spheres, fields are generated around lines or curves.
- Shape follows a skeleton.

Spline :-

A spline is a smooth curve used in computer graphics that is drawn by joining a set of points in a smooth way.

Simple ex :-

Image a flexible ruler :-

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- You fix it at some points.
- The ruler bends smoothly through those points.

That smooth bend is called a spline.

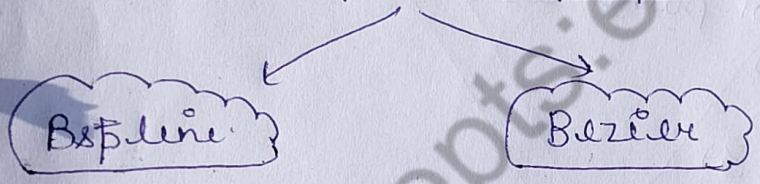
A spline is a smooth, continuous curve used in computer graphics to represent shapes, paths and surfaces.

Basic Idea :-

In early days, designers used a thin flexible strip (called a spline) to draw smooth curves by fixing it at certain points.

Computers graphics uses the same idea mathematically.

There are many types of spline :-



Bspline :-

B-spline is a basic spline.

It is a smooth curve used in computer graphics that is created using control points and basis functions.

It is an improved version of Bezier spline and is widely used in animation.

↳ The curve does not pass through all control points.

- It is more stable and flexible for complex shapes.

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B-spline curve

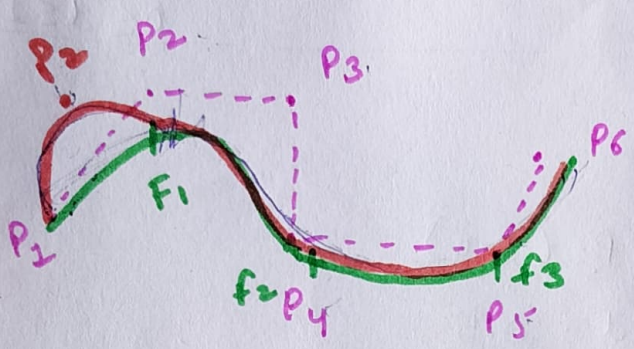
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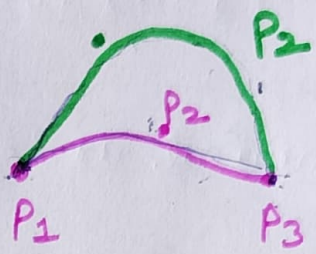
Local control

Bezier curve

↓

global control





- It is defined by control points, degree and a knot vector.
- The curve does not necessarily pass through all control points.

curve eqn :-

$$C(t) = \sum_{i=0}^{n} P_i N_{i,p}(t)$$

where

- P_i are control points
- $N_{i,p}(t)$ are B-spline functions.
- p is degree of the curve

Bezier :-

- Bezier curve is a parametric curve widely used in computer graphics for drawing smooth curves and shapes.
- It is defined by a set of control points.
- The curve starts at the first control point and ends at the last one.
- The curve always passes through the first and last control points.

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for (4 control points :- P_0, P_1, P_2, P_3).

Equation :-

$$B(t) = (1-t)^3 P_0 + 3t(1-t)^2 P_1 + 3t^2(1-t) P_2 + t^3 P_3, \quad 0 \le t \le 1$$

Numerical ex :-

let
$P_0(0,0), P_1(2,4), P_2(4,4), P_3(6,0)$

find point on curve at $t = 0.5$

$(1-0.5)^3 = 0.125$

$3(0.5)(1-0.5)^2 = 0.375$

$3(0.5)^2(1-0.5) = 0.375$

$(0.5)^3 = 0.125$

$B(0.5) = 0.125(0,0) + 0.375(2,4) + 0.375(4,4) + 0.125(6,0)$

$= 0 + 0.75 + 1.5 + 0.75, 0 + 1.5 + 1.5 + 0$

$= (3, 3)$.

So, the point on the Bezier curve at $t = 0.5$ is $\boxed{(3,3)}$ Ans.

→ all numericals will be provided on numerical video

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